VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × FLOW MEASUREMENT

HIGH-PRECISION FLOW MEASUREMENT, ANY CONDUCTIVITY

SonicLine Flowmeter
Inline Ultrasonic

A transit-time inline ultrasonic flowmeter for conductive and non-conductive liquids alike, with transducers that can be serviced under pressure and zero obstruction in the bore.



DN50–1200
NOMINAL SIZE RANGE

±0.5 %
ACCURACY OF FLOW RATE

40 bar
MAXIMUM WORKING PRESSURE

TRANSIT-TIME ULTRASONIC PRINCIPLE | **HOT-SWAP** TRANSDUCERS |
VEPOLINK FIELD-PROVEN BUILD

350 cSt	±0.5 %	200 °C	2 path
MAXIMUM LIQUID VISCOSITY	ACCURACY OF FLOW RATE	MAX. MEDIUM TEMPERATURE	STANDARD ACOUSTIC PATHS (4 OPTIONAL)

01 OVERVIEW

Flow, timed *by sound.*

An inline transit-time ultrasonic flowmeter for applications where process shutdown is not an option and where temperature or pressure extremes rule out other technologies. Unlike electromagnetic measurement, it works on any liquid under 350 cSt — conductive or not, including oils and liquefied gases.

The sensor, transducers and wall-mounted transmitter form one system. Transducers mount with O-rings or flanges and can be replaced without interrupting flow, so service never means a shutdown.

REPORTED CHANNELS

- INSTANTANEOUS FLOW RATE
- TOTALISED VOLUME
- ANALOG OUTPUT (1)
- PULSE OUTPUT (1)
- RELAY OUTPUT (1)
- HART / PROFIBUS PA
- SIGNAL QUALITY DIAGNOSTICS
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Any Conductivity

Measures all liquids under 350 cSt whether conductive or not, including oils and liquid gases that an electromagnetic meter cannot read.

Transducers Serviced Live

Hot-tap transducer fittings let technicians replace a transducer under full line pressure, with no process interruption.

No Pressure Drop

An unobstructed bore with no moving parts measures flow without adding restriction to the line.

Wall-Mounted Transmitter

The transmitter mounts on a wall or panel away from the sensor, with full self-diagnostics and standard protocol outputs.

02 MEASUREMENT PRINCIPLE

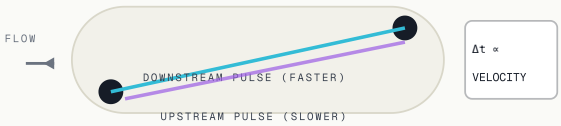
How transit-time *ultrasonic measures flow.*

A pair of transducers sends ultrasonic pulses diagonally across the pipe, one with the flow and one against it. Flow speeds up the downstream pulse and slows the upstream one — the difference in arrival time is directly proportional to velocity.

TRANSIT-TIME

Differential Time of Flight

UPSTREAM PULSE · DOWNSTREAM PULSE · ΔT



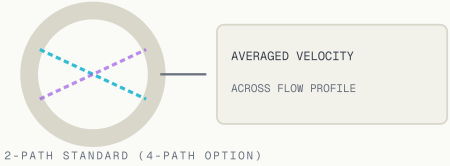
Two transducers alternate sending pulses diagonally across the flow. The downstream pulse arrives faster, the upstream pulse slower; the time difference is proportional to flow velocity.

`Δt ∝ velocity`

PROFILE

Multi-Path Averaging

2-PATH STANDARD · 4-PATH OPTION



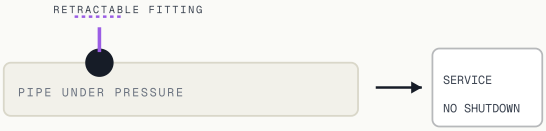
Two acoustic paths, as standard, sample the velocity profile at different depths and are averaged for accuracy across varying flow conditions; four paths are available for the most demanding installs.

`Profile-corrected velocity`

SERVICEABILITY

Live Transducer Replacement

RETRACTABLE FITTING · UNDER PRESSURE



A retractable transducer fitting lets a transducer be withdrawn and replaced without stopping the process or depressurising the line.

`Service without shutdown`

OUTPUT

Transmission & Reporting

4-20 MA · PULSE · RELAY · HART / PA



The wall-mounted transmitter linearises, displays and totalises the signal, with one analog, one pulse and one relay output alongside HART or PROFIBUS PA.

`Rate · total · diagnostics`

03 SYSTEM & CONFIGURATIONS

One system, *several duties.*

The sensor, transducers and transmitter form one measuring system, offered in configurations suited to industrial process duty, utility energy metering, and retrofit onto an existing pipe run.

CONFIG 01

Industrial

Rugged reliability for conductive and non-conductive liquids, including oils and liquid gases.

CONFIG 02

Utility

Covers district heating, chilled water and combined heating/cooling energy metering installations.

CONFIG 03

Retrofit

Installed on an existing pipe run without stopping flow — a cost-effective path to ultrasonic metering.

CONFIG 04

Special versions

Duplex/stainless sensors, ±250°C/-200°C service, 430 bar rating, and sizes down to DN 25 on request.

FLANGED OR WAFER, ANY SIZE

Supplied with or without flanges from DN 50 to DN 1200, in carbon or stainless steel from DN 100 upward, the sensor fits new installs and most existing piping standards.

1- OR 2-TRACK INTERFACES

The transmitter accepts one- or two-track sensor interfaces, so a single transmitter platform covers 2-path and higher-path sensor builds alike.

WHY THIS MEASUREMENT MATTERS

Petrochemical, power and utility plants often cannot shut a line down for sensor service and need measurement across extreme temperature and pressure ranges that rule out alternatives. Transit-time ultrasonic measurement, with live transducer service, is built for exactly that constraint.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

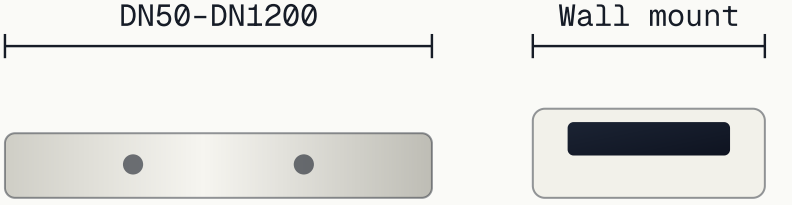
Parameter	Volumetric flow, conductive or non-conductive liquids
Working Principle	Transit-time ultrasonic
Nominal Size	DN 50 - 1200
Flow Range	Qn 10 - 4200 m³/h, depending on size
Accuracy	±0.5% of flow rate
Resolution	3 mm
Repeatability	3 mm
Working Pressure	Max. 40 bar
Working Temperature	-20 to 85 °C (transmitter)
Medium Temperature	-20 to 200 °C
Liquid Viscosity	Up to 350 cSt

SYSTEM · BUILD · POWER

Display	2×16 alphanumeric LCD with backlight
Design	2-path standard, 4-path on request
Transducer Service	Replaceable under pressure, no shutdown
Pipe Material	Steel EN 1.0345-P235GH
Transducer Material	Stainless steel AISI 316 or similar
Sensor Input	1- or 2-track interfaces
Transmitter Output	1 analog · 1 pulse · 1 relay
Transducer Output	4-20 mA · HART · PROFIBUS PA
Power Supply	24 V AC/DC or 115/230 V AC
Protection Class	IP65 (NEMA 4)
Transmitter Mounting	Wall mount only

INSTRUMENT DIMENSIONS

INDICATIVE SENSOR & WALL TRANSMITTER · NOT TO SCALE



SENSOR. DN50-1200 WALL TRANSMITTER

05 INSTALLATION & UPKEEP

Installed inline, *serviced live.*

> The sensor is installed inline, flanged or wafer-style, with the wall-mounted transmitter sited within cable reach. Transducers can be withdrawn and replaced **under full line pressure** through their retractable fitting.

01 · STRAIGHT RUNS

Upstream / downstream

Adequate straight pipe either side of the sensor settles the velocity profile the multi-path reading depends on.

02 · SENSOR ORIENTATION

Always full bore

Mount so the pipe always runs full; the acoustic paths need liquid present across their full span to read correctly.

03 · TRANSMITTER SITING

Wall-mounted

The transmitter mounts to a wall or panel within standard cable distance of the sensor, in a location accessible for service.

04 · TRANSDUCER SERVICE

Hot-tap replacement

A retractable fitting allows transducer replacement without a process shutdown or depressurising the line.

05 · CALIBRATION

Factory traceable

Each sensor ships with a factory calibration record; periodic verification follows the site's instrument audit schedule.

06 · DIAGNOSTICS

Reported with the data

Signal quality, path status and fault diagnostics travel alongside rate and totaliser outputs.

SIZING A METER

Nominal size is selected against the expected flow range (Qn 10 to 4200 m³/h, size-dependent), with path count, transducer material and special temperature or pressure ratings selected to match the process. Our team sizes the meter from your flow range and process conditions.

COMMISSIONING

Commissioning covers path signal verification, a zero check with the line full and static, and confirmation of output scaling against the plant's control, billing or energy-metering system — after which the meter runs unattended, with transducers serviceable live if ever needed.

06 APPLICATIONS

Where it *works.*

High-pressure, high-temperature and always-on lines across heavy process industries, where shutdown for sensor service is not an option.

Petrochemical Industry	Power Engineering	Water & Wastewater
Oil & Liquefied Gases	District Heating & Cooling	Energy Metering Retrofits

WHERE TRANSIT-TIME WINS OVER ELECTROMAGNETIC

Electromagnetic measurement needs a conductive liquid; transit-time ultrasonic does not. That makes this the choice for hydrocarbons, liquefied gases and other non-conductive process streams, alongside conductive utility and water duties.

RETROFIT WITHOUT DOWNTIME

Because the sensor installs inline and transducers service live, a retrofit configuration adds ultrasonic metering to an existing run without the shutdown a sensor swap would otherwise demand.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Transit-time ultrasonic. Two transducers send pulses diagonally across the flow; the difference in arrival time between the downstream and upstream pulse is proportional to velocity.

Q Does the liquid need to be electrically conductive?

No. Unlike electromagnetic flowmeters, this works on conductive and non-conductive liquids alike, including oils and liquid gases, up to 350 cSt viscosity.

Q Can transducers be serviced without a shutdown?

Yes. A retractable fitting allows transducer replacement under full line pressure, with no interruption to flow.

Q What sizes and pressures does it cover?

DN 50 to DN 1200 as standard (DN 25 on special request), up to 40 bar standard working pressure (430 bar special versions available).

Q What temperature range can it handle?

-20 to 200 °C for the process medium as standard, with special high/low temperature sensor versions to +250 °C or -200 °C on request.

Q How many acoustic paths does it use?

Two paths as standard, averaging the flow profile across the pipe; four-path sensors are available for the most demanding installations.

Q Where does the transmitter mount?

On a wall or panel, wired back to the inline sensor — the transmitter is a wall-mount unit only, not integral to the sensor.

Q What outputs are available?

One analog, one pulse and one relay output from the transmitter, with 4–20 mA, HART and PROFIBUS PA from the transducer side.

Q Is it suitable for retrofitting an existing line?

Yes — a retrofit configuration is specifically available for installing onto existing pipe runs without stopping flow.

Q What maintenance does it need?

Minimal — there are no moving parts in the flow path. Periodic signal-quality checks and, if ever needed, live transducer replacement are all that's typically required.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

FLOW, MEASURED WITHOUT A SHUTDOWN

Know your flow. *Serviced live.*

Tell us the liquid, the bore and the pressure and temperature range, and we'll configure the sensor, path count and transducer material around your process. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sizes, specifies and commissions inline ultrasonic flowmeters. No call centre.
+91 99538 86178 →

02 **Request a sizing & configuration**
RECOMMENDED
Send your bore, liquid and operating range — we'll spec path count, transducer material and output.
sales@vepolink.com →

03 **Book a demonstration**
See a running meter and the transmitter output with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM